

Customer Churn & Retention Analysis — Final Summary

Telco Customer Churn dataset (7,043 customers) — Data Analyst Internship, Task 4

Headline Numbers

Total Customers	Churned	Churn Rate	Avg. Monthly Charges	Revenue at Risk
7,043	1,869	26.54%	\$64.76	\$139,130.85 / month

Statistical Results ($\alpha = 0.05$)

Test	Result	Decision
Chi-Square: Contract vs. Churn	Chi-Square stat = 1,184.60, $p \approx 5.9\text{e-}258$	Reject H0 — associated
T-Test: Monthly Charges vs. Churn	$t = 18.41$, $p \approx 8.6\text{e-}73$ (churned avg \$74.44 vs. retained \$61.27)	Reject H0 — significant difference

Segment Needing the Most Retention Attention

New customers (tenure ≤ 12 months) — 2,186 customers, **47.0% churn rate** (highest of any segment) and **\$68,954/month revenue at risk** (also the highest of any segment, ahead of High-Value's \$62,760/month). This group is large, newly acquired, and not yet committed to a longer contract, making it the highest-leverage target for retention investment.

Key Findings

- 1 Overall churn is 26.54% (1,869 of 7,043 customers) — a material share of the customer base.
- 2 Contract length is the strongest churn driver: month-to-month churn (42.7%) is ~4x one-year (11.3%) and ~15x two-year (2.8%); confirmed statistically significant (Chi-Square test, $p < 0.001$).
- 3 Churn risk is front-loaded by tenure: first-year customers churn at 47.4% vs. 9.5% for customers with 49+ months tenure.
- 4 Churned customers pay more, not less — \$74.44/month average vs. \$61.27/month for retained customers, a statistically significant gap (t-test, $p < 0.001$).
- 5 Manual payment and missing services compound risk: Electronic check users churn at 45.3% (vs. ~15-17% for automatic payments), and customers without Tech Support churn at 41.6% (vs. 15.2% with it).

Recommendations

- 1 Incentivize contract upgrades — offer a discount for month-to-month customers who switch to a one-year contract, directly targeting the strongest churn driver found.
- 2 Build a structured first-year onboarding/retention program — front-load engagement (check-ins, early offers, proactive support) during the highest-risk 0-12 month tenure window.
- 3 Review fiber/high-tier pricing and value communication — test bundling a retention perk with the priciest plans, since churned customers pay \$13/month more on average and Fiber churns at 41.9%.
- 4 Push electronic check users onto automatic payment — offer a small discount for switching to bank transfer or credit card autopay, cutting churn from a method carrying 3x the risk.
- 5 Bundle a free Tech Support trial into first-year onboarding for New-segment customers — this segment carries the highest churn rate and the most revenue at risk of any segment.

Full analysis, charts, and statistical test code: Task4_Customer_Churn_Analysis.ipynb. Dashboard data and build guide: power_bi/ folder.